

STANDARD QUALITY ASSURANCE PLAN FOR ELASTOMERIC BEARINGS

S. No.	Component Operation	Characteristic Checks & Tolerances	Frequency & Type of check	Reference Document	Fabricator's quality control	Inspection details		Type of Records	Acceptance criteria	Remarks
						Inspecting Agency	Extent of Inspection			
1	2	3	4	5	6	7	8	9	10	11
1.0	RAW MATERIAL Elastomer	Physical and Chemical Test	As per manufacturer certificate & test required by Authorized inspecting Official of Zonal Railway from Govt./ NABL approved lab	Manufacturer test certificate & Challan	Verification of Reference document.	Authorized inspecting Official from Zonal Railway	From batches at Random and at discretion of Inspecting official	Fabricators record.	IRC:83 (Part-II) & referred documents	- The elastomer used in the manufacture of Elastomeric Bearing should be either Natural Rubber (NR) or Chloroprene Rubber (CR) as the raw polymer. - Raw material must meet requirements given in Para 4 (Material) of IRC:83 (Part-II).
2.0	Mild Steel	<u>Mechanical & Chemical Test</u> UTS, Yield Stress, Elongation%, Bend test, Impact test at 0 degree centigrade or -20 degree for sub-zero zone, test for C, Mn, Si, P, S etc.	As per Mill TC & test required by Authorized inspecting official of Zonal Railway from Govt. Lab./ NABL approved lab.	Mill TC and Challan	Verification of Reference document and Complete visual inspection	Authorized Inspecting official of Zonal Railway	One test piece per thickness per heat. / At discretion of Inspecting official	Fabricators record Co-related with Mill TC & test report.	IS: 2062, E250, Quality BO is generally used. However Quality C shall be used for subzero condition. (As specified in approved Drg.).	- Laminates of mild steel conforming to IS: 2062 / IS: 1079 or equivalent international grade shall be used. The yield stress of the material shall not be lesser than 250 Mpa. - Uses of any other materials like fibreglass or similar fabric as laminates are not permitted for the purpose of this Code. - Physical condition of raw material must be visually checked for Pitting, rusting, rolling defect etc before use. - Corrosion protection of exposed steel surfaces, if any, shall conform to the requirement of



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										relevant part covering corrosion protection of bearing plates. 5. 100 % Ultra Sonic Testing shall be conducted for 12mm & above thick plates as per provision of relevant specification (IS:4225 or ASTM A435) before processing the material for manufacturing.
3.0	<u>Manufacturing Process</u>	Dimension and tolerances.	Visual & Measurement	As per approved drawing	Measurement of dimension	Manufacturer inspecting agency	At Discretion of Inspecting official	Fabricators record.	IRC 83 Part-II & as per approved drawing	1 All Bearing shall conform to the provisions in Clause 6 on Manufacturing and Workmanship and Table 5: Manufacturing Tolerances of IRC:83 (Part-II) in all respects. 2 Internal report shall be furnished which will be verified by Zonal Railway Inspector. However random check may be done by Zonal Railway inspector during Manufacturing. 3 Manufacturing process (any stage) shall be open to inspection by Railway inspecting official.



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4.0	Final inspection (See Note below)									
4.1	General Inspection	Dimensions of bearing as per specified tolerances.	Visual examination and dimensional check	As per relevant drg. & Specification	Measurement of dimension	Authorized inspecting officials from Zonal Railway	100%	Fabricators record & test report.	IRC 83 (Part-II) & referred documents	1. All Bearing shall conform to the provisions in Clause 7.9.1.1 of IRC:83 (Part-II) in all respects for Level 1 acceptance testing. 2. All Bearing shall conform to the provisions in Clause 7.9.2.1 of IRC:83 (Part-II) in all respects for Level 2 acceptance testing.
4.2	Chemical & Physical Test	Physical and Chemical Test.	As per manufacturer certificate & test required by Authorized inspecting Official of Zonal Railway from Govt./ NABL approved lab	Manufacturer Test Certificate & Challan	Verification of reference document.	Authorized Inspecting Official from Zonal Railway	From batches at Random/ Discretion of Inspecting official	Fabricators record.	IRC 83 (Part-II) & referred documents	1. Specially moulded test pieces shall conform to the provisions in Clause 7.9.1.2 of IRC: 83 (Part-II) in all respects for Level 1 acceptance testing/ Level 2 acceptance testing. 2. Ozone resistance test shall be mandatory for all acceptance level. 3. Test certificate shall be furnished by manufacturers for level I acceptance/ Level II acceptance.
4.3	Load Test on complete bearings									
a)	Axial load test	Variation in stiffness	All bearing of the lot shall be tested.	As per relevant drg. and specification	Fabricators record.	Authorized Inspecting Official from Zonal Railway	100% In case of large lot	Fabricators record.	IRC 83 (Part-II) & referred documents	For both Acceptance Level 1 & Acceptance Level 2, all 100% bearings to be checked for axial load tests as per Clause 7.9.1.1 (C) of IRC:83 (Part-II)
b)	Shear Modulus 	Shear modulus by plotting shear stress strain curve	Load testing on test bearings.	As per relevant drg. and specification	Fabricators record.	Authorized Inspecting Official from Zonal Railway	Mandatory for Large lot and on discretion of Inspecting	Fabricators record.	IRC 83 (Part-II) & referred documents	Bearing to be checked as per Clause 7.9.2.3 of IRC:83 (Part-II)

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							official for small lot			
c)	Compressive Stiffness Test	Compressive load deformation curve and compressive stiffness	Load testing on test bearings.	As per relevant drg. and specification	Fabricators record.	Authorized Inspecting Official from Zonal Railway	Mandatory for Large lot and on discretion of Inspecting official for small lot	Fabricators record.	IRC 83 (Part-II) & referred documents	Bearing to be checked as per Clause 7.9.2.3 of IRC:83 (Part-II)
d)	Shear Bond Test	Required adhesion between elastomer to steel laminate.	Load testing on test bearings for.	As per relevant drg. and specification	Fabricators record.	Authorized Inspecting Official from Zonal Railway	Mandatory for Large lot and on discretion of Inspecting official for small lot	Fabricators record.	IRC 83 (Part-II) & referred documents	Bearing to be checked as per Clause 7.9.2.3 of IRC:83 (Part-II)
4.4	Special Acceptance Inspection									
	TGA Method /Any Other Test as per Special Contract Agreement/ As per provisions in Clause 7.9.3 of IRC: 83 (Part-II)	Special type of test (Other than test mentioned in clause 4.3) 	As per manufacturer certificate & test required by Authorized inspecting Official of Zonal Railway from Govt./ NABL approved lab	As per special contract agreement	Fabricators record.	Authorized Inspecting Official from Zonal Railway	Discretion of Inspecting official	Test report from Govt./ NABL approved lab	IRC 83 (Part-II) & referred documents	As per provisions in Clause 7.9.3 of IRC: 83 (Part-II)

Note:

- For the purpose acceptance testing (Clause 7.9 of IRC:83- Part II), acceptance lots shall be classified as below:
A lot size of 24 or larger number of bearings shall be defined as a large lot and a lot size of less than 24 bearings shall be defined as small lot.
When the number of bearings for a project is large and phased production is permitted, bearing supplied in any one phase will be considered as a large lot.
- The level of acceptance testing shall generally be graded into the following two levels depending on lot size:
Level 1 acceptance testing
Level 2 acceptance testing
Acceptance testing level 1 is a higher-level inspection and shall be applicable to large lots. This level of inspection can be specified for small lots also as per the discretion of the Railway. This shall involve manufacture of extra bearings to be used as test bearing and eventually consumed in destructive testing. Acceptance testing level 2 shall be applicable in general for all lots, unless otherwise specified, and shall not involve any destructive testing of finished bearings.